

Significant figures

- Significant figures are the meaningful digits in a measured or calculated quantity.

Rules of significant figures

1. Any digit that is not zero is significant. Thus, 845 cm has three significant figures, 1.234 kg has four significant figures, and so on.
2. Zeros between nonzero digits are significant. Thus, 606 m contains three significant figures, 40,501 kg contains five significant figures, and so on.
3. Zeros to the left of the first nonzero digit are not significant. Their purpose is to indicate the placement of the decimal point. For example, 0.08 L contains one significant figure, 0.0000349 g contains three significant figures, and so on.

4. If a number is greater than 1, then all the zeros written to the right of the decimal point count as significant figures. Thus, 2.0 mg has two significant figures, 40.062 mL has five significant figures, and 3.040 dm has four significant figures. If a number is less than 1, then only the zeros that are at the end of the number and the zeros that are between nonzero digits are significant. This means that 0.090 kg has two significant figures, 0.3005 L has four significant figures, 0.00420 min has three significant figures, and so on.

5. For numbers that do not contain decimal points, the trailing zeros (that is, zeros after the last nonzero digit) may or may not be significant. Thus, 400 cm may have one significant figure (the digit 4), two significant figures (40), or three significant figures (400). We cannot know which is correct without more information. By using scientific notation, however, we avoid this ambiguity. In this particular case, we can express the number 400 as 4×10^2 for one significant figure, 4.0×10^2 for two significant figures, or 4.00×10^2 for three significant figures.

Rounding-off

Always round to the **even** number, if the **last digit is 5**

$$8.\textcolor{red}{6}5 = 8.\textcolor{red}{6}$$

$$8.\textcolor{red}{7}5 = 8.\textcolor{red}{8}$$

$$8.\textcolor{red}{5}5 = 8.\textcolor{red}{6}$$

Rounding-off

1. If the digit is less than 5, simply drop it and all digits farther to the right.

Example:

8.724 rounds off to 8.72 if we want only two digits after the decimal point

Rounding 1.2143 to three significant figures gives 1.21

2. If the first digit following the point of rounding off is greater than 5, we add 1 to the preceding digit.

Example:

8.727 rounding off to 8.73

0.425 rounds off to 0.42

Rounding off 1.2151 to three significant figures gives 1.22

$$9.47 = 9.5$$